

SkelGym Multi-Stream Kinematic Fusion Hierarchy & Dual-Stage Decision Engine

From Orthogonal Skeleton Streams to Cross-Paradigm SLSQP Ensembles and Video-Consensus Soft Pooling

LEVEL 1: Orthogonal Stream Generation from MediaPipe Pose (T=32 Frames, V=13 Load-Bearing Joints)

Stream 1: Joint Position (Rel 3D)

$X_{\text{joint}}(c,t,v) = \tilde{p}(t, v) \in \mathbb{R}^3 \times 32 \times 13$
Pelvis-centered relative 3D coords
Encodes absolute spatial posture

Stream 2: Bone Vector (Bone 3D)

$X_{\text{bone}}(c,t,v) = \tilde{p}(\text{target}) - \tilde{p}(\text{source})$
Directed limb segment vectors
Spatial variance & length invariant

Stream 3: Joint Motion (J-Mot)

$X_{\text{jmot}}(c,t,v) = X_{\text{joint}}(t+1) - X_{\text{joint}}(t)$
First-order joint velocity (Δp)
Captures instantaneous limb speed

Stream 4: Bone Motion (B-Mot)

$X_{\text{bmot}}(c,t,v) = X_{\text{bone}}(t+1) - X_{\text{bone}}(t)$
Rate of change of bone vectors (Δb)
Encodes rotational joint velocity

Stream 5: Sequence Mix (117-d)

$X_{\text{mix}} \in \mathbb{R}^{32 \times 117}$
39 Joints $\times$ 3 Coordinates
Global spatio-temporal flattened stream

LEVEL 2: Single-Stream Expert Models (Trained from Scratch, ~350K-400K Parameters Each)

AAGCN Joint Expert

Adaptive Graph Conv (V=13)
Window Acc: 61.72% | Video: 69.96%
Params: 378K | FLOPs: 101.4M

AAGCN Bone Expert (Strongest Graph)

Adaptive Graph Conv (V=13)
Window Acc: 65.84% | Video: 72.96%
Params: 378K | FLOPs: 101.4M

AAGCN Joint-Motion Expert

Adaptive Graph Conv (V=13)
Window Acc: 46.81% | Video: 64.38%
Params: 378K | FLOPs: 101.4M

AAGCN Bone-Motion Expert

Adaptive Graph Conv (V=13)
Window Acc: 57.49% | Video: 79.83%
Params: 378K | FLOPs: 101.4M

Transformer Mix Expert

Pre-LN MHSA (4L, 8H, d=128)
Window Acc: 64.53% | Video: 72.39%
Params: 399K | FLOPs: 12.5M

LEVEL 3: Hierarchical Multi-Stream Fusion Paradigms & Benchmark Configurations

Config 1: Two-Stream AAGCN

Static Kinematics Late Fusion

Streams: Joint (3D) + Bone (3D)

- Combines position & segment angles
- Window Acc: 66.61% | Video: 73.82%
- Params: 756K | FLOPs: 202.8 MFLOPs
 - Host CPU Latency: 1.97 ms

Config 2: Four-Stream AAGCN

Complete Unified Graph Paradigm

Streams: Joint + Bone + J-Mot + B-Mot

- Spatial posture + dynamic velocities
 - Window Acc: 68.78% $\pm$ 1.35%
- Video Acc: 77.25% $\pm$ 1.55% (F1: 0.7678)
- Params: 1.51M | FLOPs: 405.7 MFLOPs
 - Host CPU Latency: 3.84 ms

★ Config 3: SkelGym-Lite (Edge-Optimal)

Cross-Paradigm Minimal Hybrid Ensemble

Streams: Transformer Mix + AAGCN Bone

- Global Self-Attention + Local Bone Topology
 - Window Acc: 69.66% $\pm$ 0.71%
- Video Acc: 76.68% $\pm$ 0.25% (F1: 0.7582)
 - Params: 777K | FLOPs: 113.93 MFLOPs
- Host CPU Latency: 1.44 ms (CUDA: 0.19 ms)
- Optimal Pareto efficiency for mobile/edge!

★ Config 4: SkelGym-Full (SOTA Ensemble)

5-Stream Cross-Paradigm Late Fusion

Streams: Transformer Mix + 4-Stream AAGCN

- Unified Sequence & 4-Stream Graph Experts
- Window Acc: 70.03% $\pm$ 0.70% (F1: 0.6877)
 - Video Acc: 77.68% $\pm$ 0.86% (F1: 0.7689)
- Baseline Seed Video Acc: 77.68% (F1: 0.7649)
 - Params: 1.91M | FLOPs: 418.21 MFLOPs
- Host CPU Latency: 4.33 ms (CUDA: 0.54 ms)

LEVEL 4: Validation-Calibrated SLSQP Simplex Optimization & Dual-Stage Decision Engine

SLSQP Constrained Optimization

Calibrated strictly on Validation Partition (Zero Leakage)

Minimizes Negative Log-Likelihood (NLL):
 $w^* = \underset{w}{\operatorname{argmin}} \sum_{i=1}^N \log(\sum_{m=1}^M w_m \cdot P_{\{m, y_i\}})$
subject to: $\sum w_m = 1, w_m \geq 0$ (Probability Simplex Δ^4)

- Empirical SLSQP Stream Allocation:
- AAGCN Bone: 62.38% $\pm$ 11.58% (Primary Driver)
 - Transformer Mix: 26.58% $\pm$ 6.51% (Sequence Complement)
 - Bone-Motion: 14.54% | Joint-Motion: 2.86% | Joint: ~0%

Stage 1: Window-Level Weighted Soft Voting

$$P_{\text{win}}(w) = \sum_{m=1}^5 w_m \cdot P_m(w)$$

- Evaluated per 32-frame sliding window (1.07s)
- Captures fine-grained instantaneous phase kinetics
- Resolves rapid intra-repetition motion transitions
- Window Accuracy: 70.03% $\pm$ 0.70% (F1: 0.6877)

Stage 2: Temporal Video Consensus Aggregation

Temporal Soft-Pooling over K_v sliding windows:
$$\hat{P}_{\text{vid}}(c) = \sum_{m=1}^5 w_{\{\text{vid}, m\}} * [(1/K_v) \sum_{k=1}^{K_v} P_{\{m,c\}}(w_{\{v,k\}})]$$

$$\hat{y}_{\text{vid}} = \underset{c}{\operatorname{argmax}} \hat{P}_{\text{vid}}(c) \quad \text{over 22 Resistance Exercises}$$

- Biomechanical & Statistical Benefits:
- Low-pass expectation filter attenuates tracking noise & occlusion spikes
 - Cancels out non-cyclic boundary states (un-racking weights, pauses)
 - Yields consistent +7.65% surge over window accuracy!
 - Video Consensus Accuracy: 77.68% $\pm$ 0.86% (F1: 0.7689)